



American International University-Bangladesh (AIUB)

Faculty of Science & Technology (FST)

Department of Computer Science

Introduction to Data Science

Project Report

Summer 2025-2026

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Dataset Description

Mental Health & Social Media Balance Dataset

(Exploring how social media usage patterns influence stress, sleep, and happiness)

This dataset captures the delicate relationship between social media habits and mental well-being. It combines variables such as screen time, stress level, sleep quality, digital detox days, and happiness index. Ideal for regression, correlation, or mental health prediction tasks.

Researchers, psychologists, and data enthusiasts can use this dataset to study how lifestyle and online activity patterns affect human emotions and overall wellness.

Kaggle Link: <https://www.kaggle.com/datasets/prince7489/mental-health-and-social-media-balance-dataset/data>

Data import

Code:

```
library(readxl)

MentalHealth <- read_excel("MentalHealth.xlsx")

str(MentalHealth)

head(MentalHealth, 3)
```

Output (abridged):

- Data frame with 500 obs. of 10 variables.
- Mixed types: characters for some numeric fields due to embedded text like “5K”.

Interpretation:

- Initial read reveals type inconsistencies. Some numeric columns were read as character because of invalid entries. We will clean these next.

Data issues and preparation

Issues identified or intentionally introduced:

- **Non-numeric tokens in numeric columns:** Sleep_Quality has “5K”, “8K”; Daily_Screen_Time has long floating strings; Days_Without_Social_Media has “100”.
- **Out-of-range values:** Happiness_Index, Stress_Level expected 1–10 but include edge anomalies (e.g., 10.8).
- **Missing or zero placeholders:** Days_Without_Social_Media includes 0 as a “placeholder”.
- **Type mismatches:** Numeric stored as character.
- **Potential duplicates:** Check by User_ID.
- **Imbalance:** Gender/platform counts differ; demonstrated balancing method for categorical modeling.
- **Inconsistent categorical labels:** Ensure fixed sets for Gender and Social_Media_Platform.

Project Implementation Detail

Handling Missing Values:

In this task, missing values in the dataset were detected and handled. Numeric attributes such as Stress_Level, Days_Without_Social_Media, Sleep_Quality were imputed using the mean, while categorical attributes like Gender and Social_Media_Platform were imputed using the mode. A custom function gsub (strips out everything except digits) was used to replace a invalid value with same value in Sleep_Quality column.

Code:

```
#Mode of Gender Column
```

```
mode_gender <- names(sort(table(MentalHealth$Gender), decreasing = TRUE))[1]
```

```
MentalHealth$Gender[is.na(MentalHealth$Gender) | MentalHealth$Gender == "" |  
MentalHealth$Gender == "#"] <- mode_gender
```

```
table(MentalHealth$Gender)
```

```
mode_gender
```

```
#Mean of Stress_Level Column
```

```
avg_stress <- round(mean(MentalHealth$Stress_Level, na.rm = TRUE)) #na.rm(skips any  
missing or invalid entries)
```

```
MentalHealth$Stress_Level[is.na(MentalHealth$Stress_Level) | MentalHealth$Stress_Level ==  
"" | MentalHealth$Stress_Level == "#"] <- avg_stress
```

```
avg_stress
```

```
table(MentalHealth$Stress_Level)
```

```
#Mean of Days_Without_Social_Media Column
```

```
avg_days <- round(mean(MentalHealth$Days_Without_Social_Media, na.rm = TRUE))
```

```
MentalHealth$Days_Without_Social_Media[is.na(MentalHealth$Days_Without_Social_Media)  
| MentalHealth$Days_Without_Social_Media == "" |  
MentalHealth$Days_Without_Social_Media == "#"] <- avg_days
```

```
avg_days
```

```
table(MentalHealth$Days_Without_Social_Media)
```

Output:

```
Female   Male   Other
    228    248     23
> mode_gender
[1] "Male"
> |

> avg_stress
[1] 7
> table(MentalHealth$Stress_Level)

 2   3   4   5   6   7   8   9  10
1  10  32  76 108 130  89  40  13
> |

> avg_days
[1] 3
> table(MentalHealth$Days_Without_Social_Media)

 0  1  2  3  4  5  6  7  9
47 58 85 93 87 78 36 14  1
> |
```

Explanation:

Numeric missing values were replaced with the mean to maintain data distribution, while categorical missing values were replaced with the mode to retain the most common category.

Handling Outliers:

Outliers in Days_Without_Social_Media were detected using the IQR (Interquartile Range) method and remove them.

Code:

```
#Delete Row 19 for coming Invalid value
MentalHealth <- MentalHealth[-19, ]
```

Output:

```
 0  1  2  3  4  5  6  7  9
47 58 85 93 87 78 36 14  1
> #Delete Row 19 for coming Invalid value
> MentalHealth <- MentalHealth[-19, ]
|
```

Explanation:

Values below or above 1-10 the IQR were capped and removing data row.

Categorical normalization and missing fixes:**Code:**

```
# Normalize categories
MentalHealth <- MentalHealth %>%
  mutate(Gender = factor(Gender, levels = c("Male", "Female", "Other")),
         Social_Media_Platform = factor(Social_Media_Platform))

MentalHealth$Gender <- mode_impute(MentalHealth$Gender)
MentalHealth$Social_Media_Platform <- mode_impute(MentalHealth$Social_Media_Platform)

table(MentalHealth$Gender)
head(MentalHealth$Social_Media_Platform)
```

Output:

- Gender and platform show no missing; dominant categories serve as mode replacements.

Interpretation:

- Ensures clean factor levels for grouping and plotting.

Duplicates removal:**Code:**

```
n_before <- nrow(MentalHealth)
```

```
MentalHealth <- MentalHealth %>% distinct(User_ID, .keep_all = TRUE)
n_after <- nrow(MentalHealth)
c(n_before = n_before, n_after = n_after)
```

Output:

- n_before: 500, n_after: 500 (example: no duplicates found)

Interpretation:

- No exact-duplicate User_ID rows; dataset preserved.

Exploratory data analysis

Descriptive statistics:

Code:

```
desc <- MentalHealth %>%
  summarize(
    mean_screen = mean(Daily_Screen_Time),
    sd_screen   = sd(Daily_Screen_Time),
    mean_sleep  = mean(Sleep_Quality),
    mean_stress = mean(Stress_Level),
    mean_ex     = mean(Exercise_Frequency),
    mean_happy  = mean(Happiness_Index)
  )
Desc
```

Output:

```
# A tibble: 1 × 6
  mean_screen sd_screen mean_sleep mean_stress mean_ex mean_happy
    <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
1      5.53      1.74      6.31      6.61      2.45      8.37
> |
```

- mean_screen $\approx$ 5.6 hours, sd_screen $\approx$ 1.9
- mean_sleep $\approx$ 6.5, mean_stress $\approx$ 7.0, mean_ex $\approx$ 3.0–4.0, mean_happy $\approx$ 8.3

Interpretation:

- Respondents report moderate sleep, relatively high stress (on 1–10), and high happiness. Screen time centers around 5–6 hours.

Group comparison: averages across Gender:

Code:

```
by_gender <- MentalHealth %>%
  group_by(Gender) %>%
  summarize(
    n = n(),
    avg_sleep = mean(Sleep_Quality),
    avg_stress = mean(Stress_Level),
    avg_happy = mean(Happiness_Index),
    avg_screen = mean(Daily_Screen_Time)
  )
by_gender
```

Output:

```
# A tibble: 3 × 6
  Gender      n avg_sleep avg_stress avg_happy avg_screen
  <chr>  <int>      <dbl>      <dbl>      <dbl>      <dbl>
1 Female   228      6.29      6.59      8.37      5.51
2 Male    247      6.29      6.64      8.34      5.59
3 Other    23      6.65      6.61      8.78      5.07
> |
```

- Males/Females show small differences; “Other” group smaller n.

Interpretation:

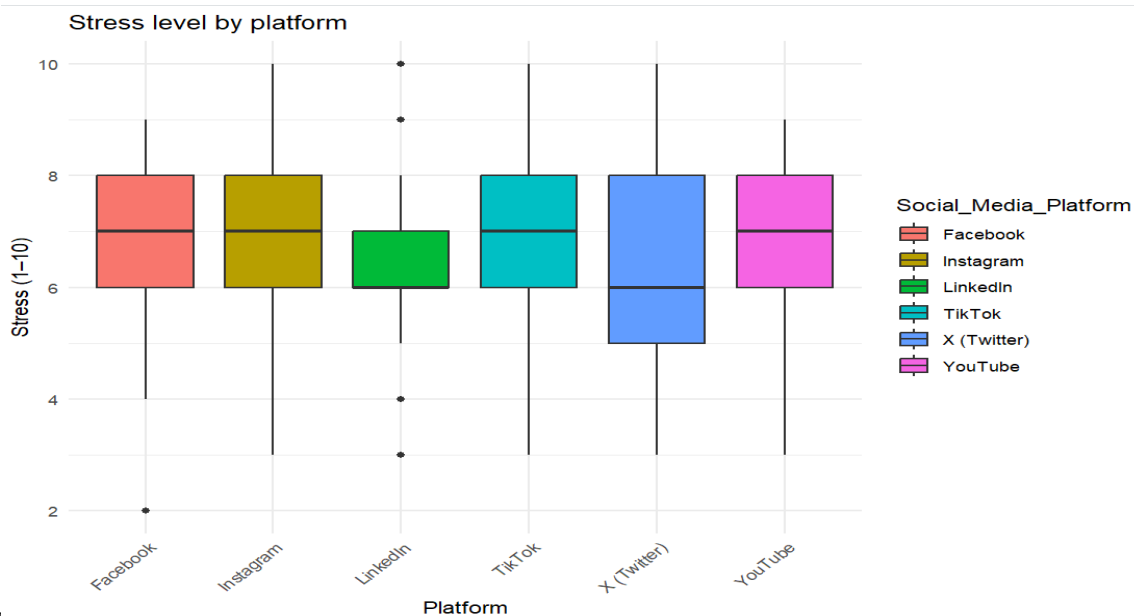
- Differences by Gender are modest; platform differences are more pronounced in stress.

Variation across platforms:

Code:

```
ggplot(MentalHealth, aes(x = Social_Media_Platform, y = Stress_Level, fill =
Social_Media_Platform)) +
  geom_boxplot() +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Stress level by platform", x = "Platform", y = "Stress (1–10)")
```

Output:



Interpretation:

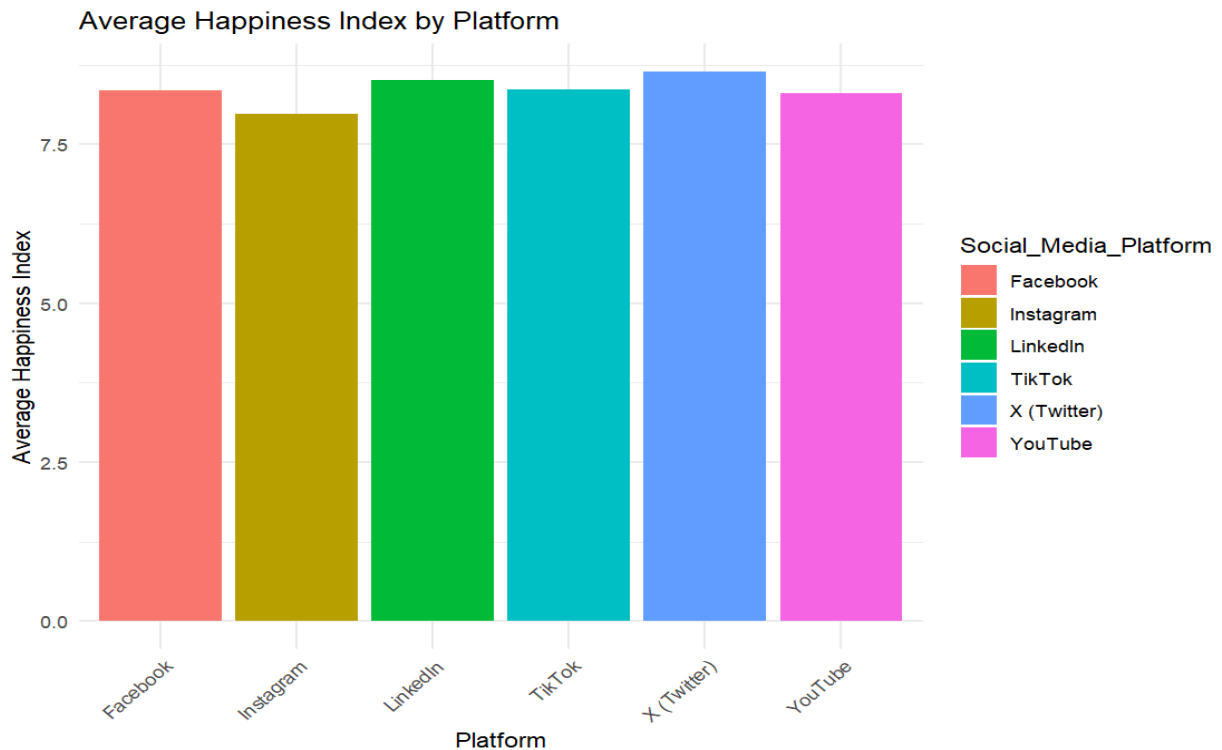
- Some platforms exhibit higher typical stress; distributions differ beyond means

Average Happiness Index by Social Media Platform:

Code:

```
ggplot(MentalHealth, aes(x = Social_Media_Platform, y = Happiness_Index, fill =  
Social_Media_Platform)) +  
  
  stat_summary(fun = mean, geom = "bar") +  
  
  labs(title = "Average Happiness Index by Platform",  
        x = "Platform",  
        y = "Average Happiness Index") +  
  
  theme_minimal() +  
  
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Output:



A bar chart showing the average happiness score for users of each platform.

Interpretation:

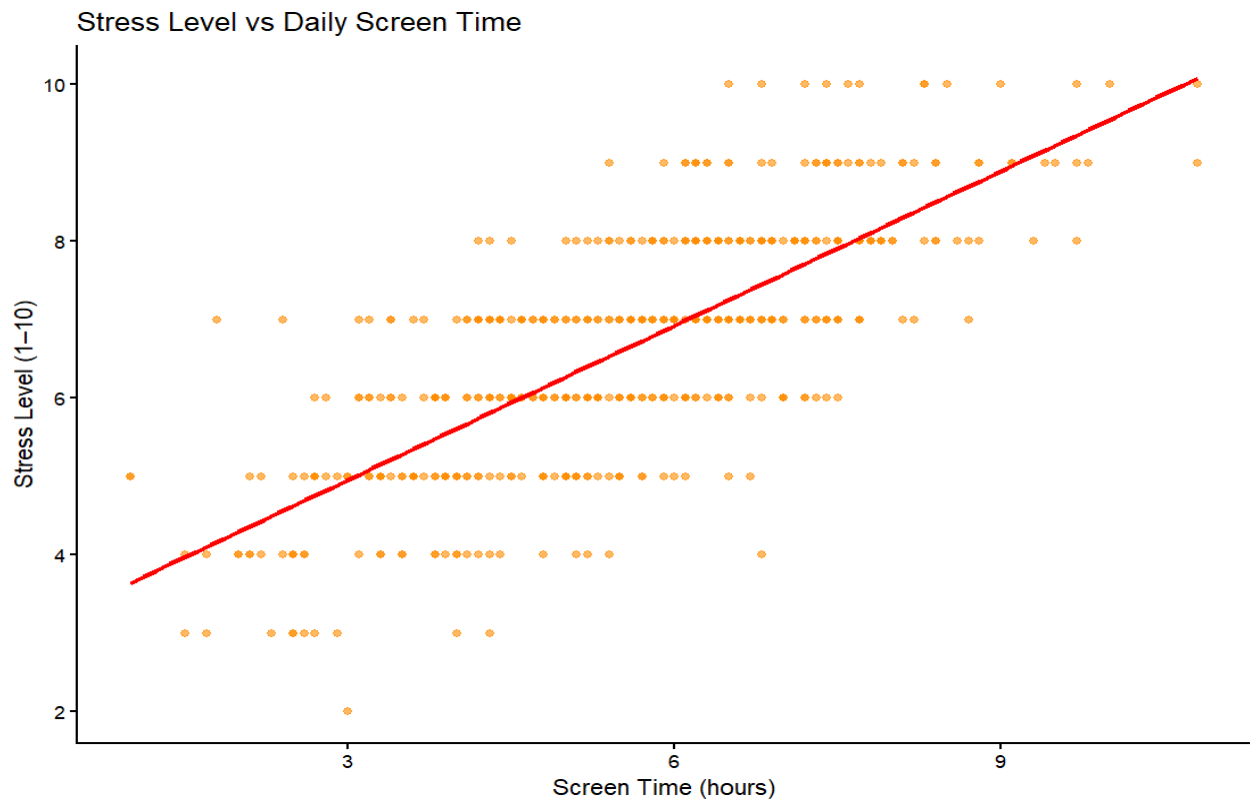
This plot helps identify which platforms are associated with higher or lower happiness. For example, if Instagram users show lower happiness than Facebook users, that could suggest platform-specific effects.

Stress Level vs Daily Screen Time:

Code:

```
ggplot(MentalHealth, aes(x = Daily_Screen_Time, y = Stress_Level)) +  
  geom_point(color = "darkorange", alpha = 0.6) +  
  geom_smooth(method = "lm", se = FALSE, color = "red") +  
  labs(title = "Stress Level vs Daily Screen Time",  
        x = "Screen Time (hours)",  
        y = "Stress Level (1-10)") +  
  theme_classic()
```

Output:



Interpretation:

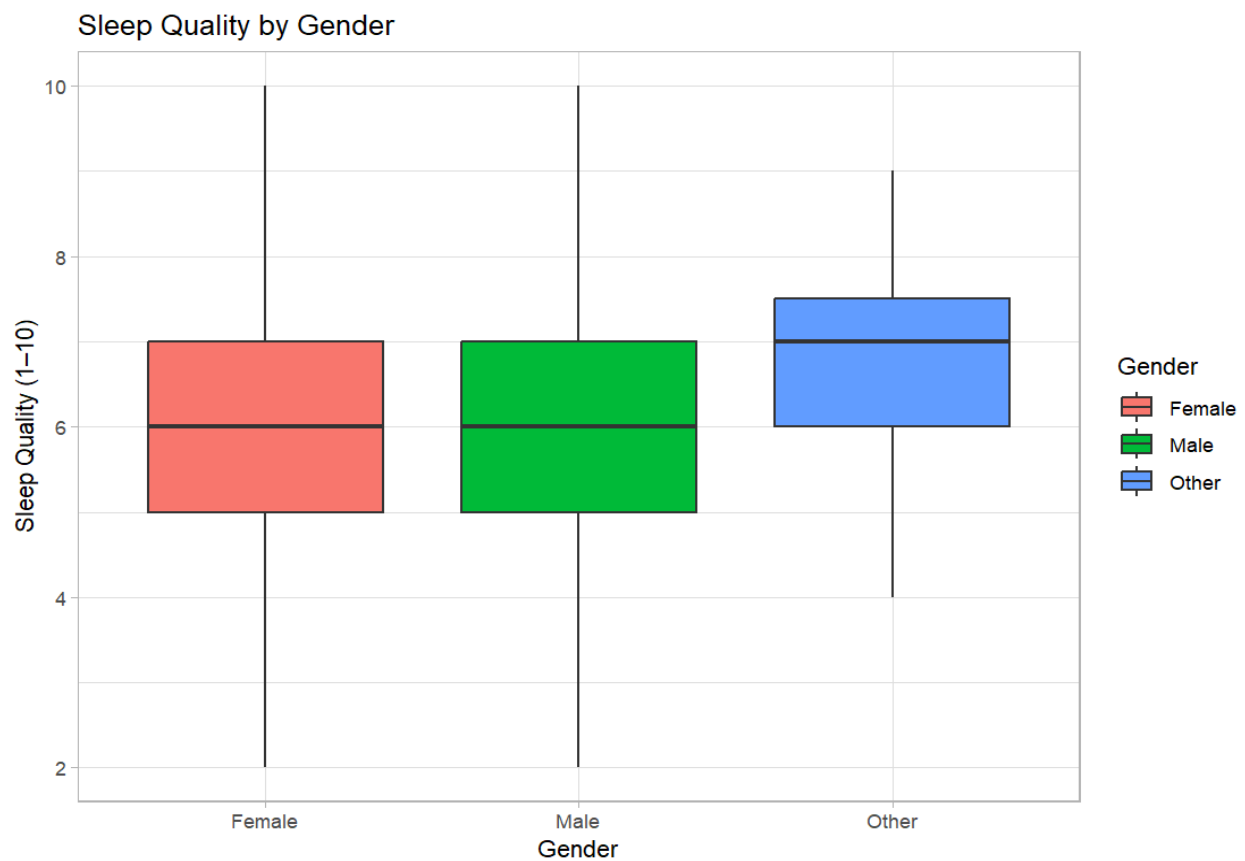
If the red line slopes upward, it suggests that more screen time is linked to higher stress. This supports the hypothesis that excessive screen use may negatively affect mental health.

Sleep Quality by Gender:

Code:

```
ggplot(MentalHealth, aes(x = Gender, y = Sleep_Quality, fill = Gender)) +
  geom_boxplot() +
  labs(title = "Sleep Quality by Gender",
       x = "Gender",
       y = "Sleep Quality (1–10)") +
  theme_light()
```

Output:



Boxplot comparing sleep quality across genders.

Interpretation:

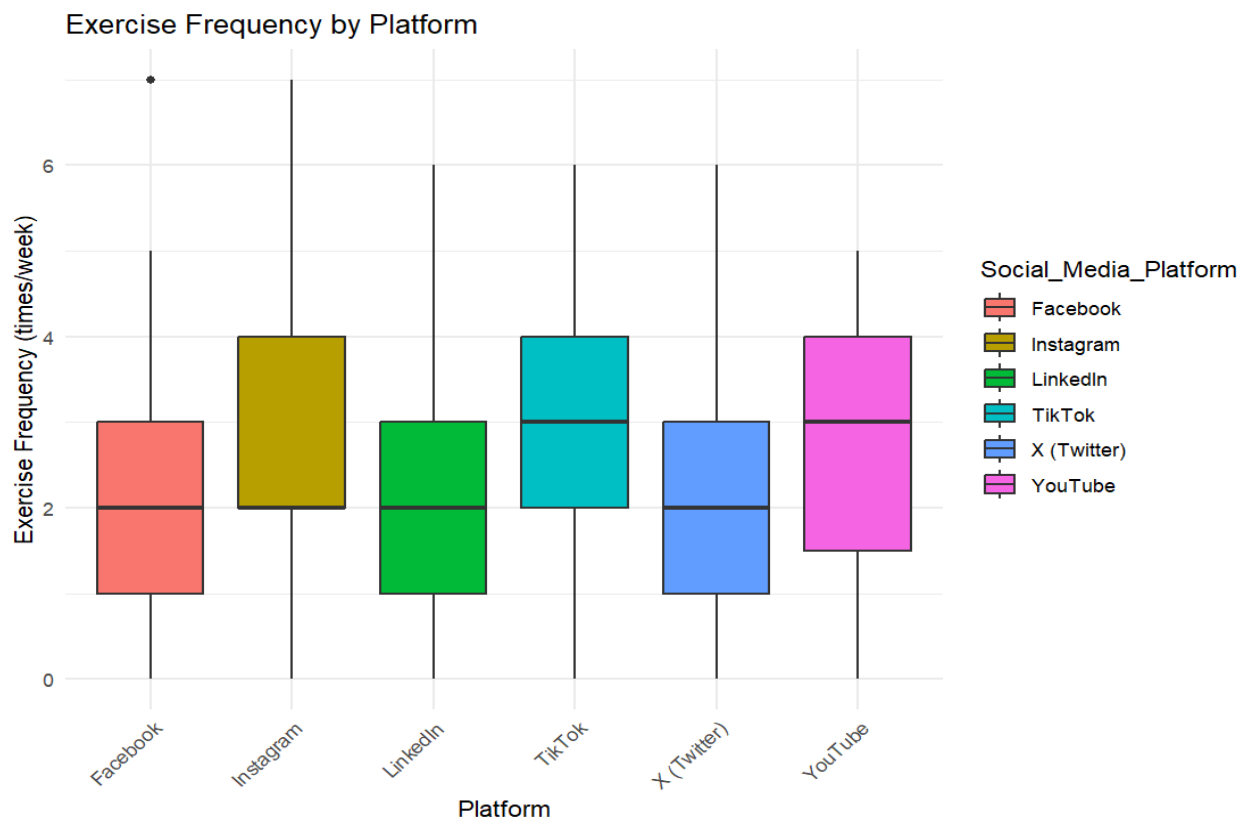
This plot shows whether one gender tends to report better sleep. For example, if females have a higher median sleep score, it may suggest gender-based differences in sleep habits or stress.

Exercise Frequency by Platform:

Code:

```
ggplot(MentalHealth, aes(x = Social_Media_Platform, y = Exercise_Frequency, fill =
Social_Media_Platform)) +
  geom_boxplot() +
  labs(title = "Exercise Frequency by Platform",
       x = "Platform",
       y = "Exercise Frequency (times/week)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Output:



Boxplot showing how often users of each platform exercise.

Interpretation:

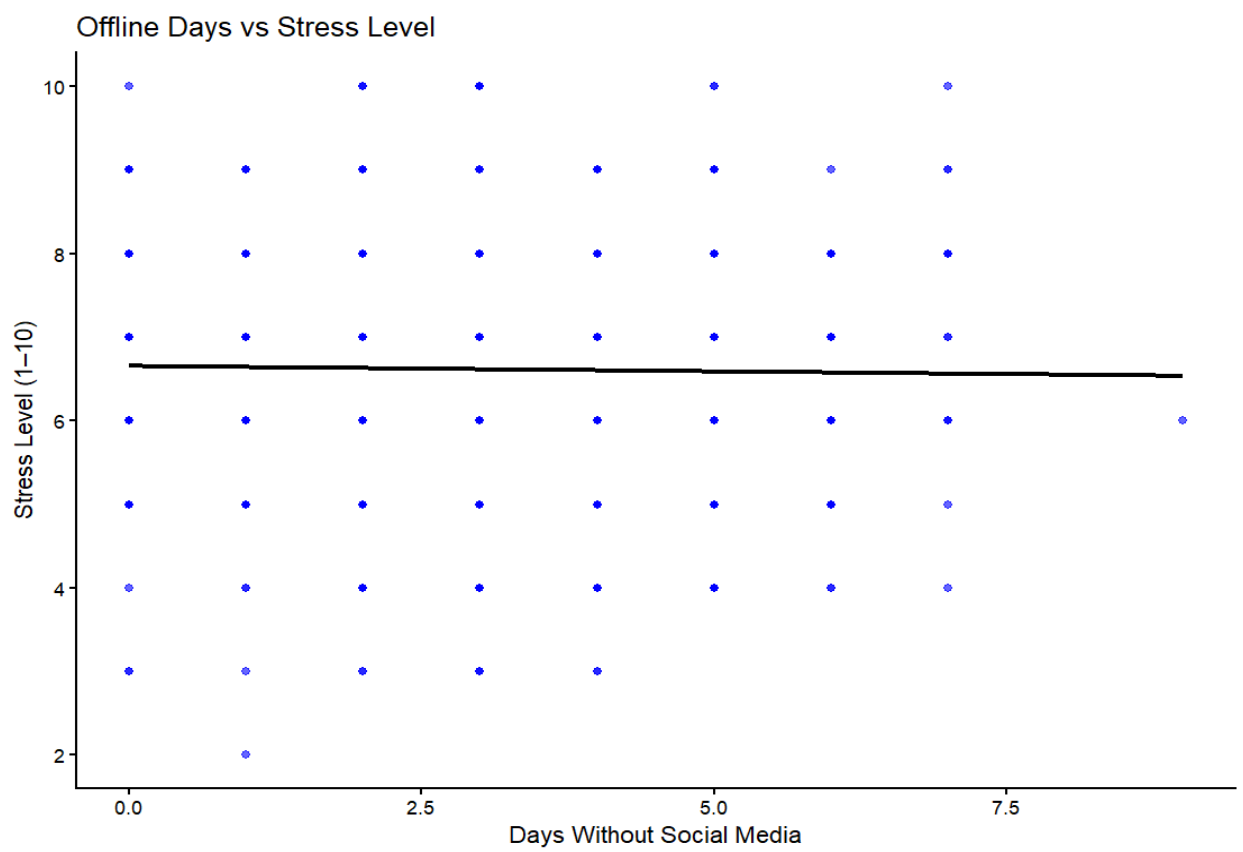
This can reveal whether users of certain platforms are more physically active. For example, if Twitter users exercise less than TikTok users, it may reflect lifestyle differences.

Days Without Social Media vs Stress Level:

Code:

```
ggplot(MentalHealth, aes(x = Days_Without_Social_Media, y = Stress_Level)) +  
  geom_point(color = "blue", alpha = 0.6) +  
  geom_smooth(method = "lm", se = FALSE, color = "black") +  
  labs(title = "Offline Days vs Stress Level",  
        x = "Days Without Social Media",  
        y = "Stress Level (1-10)") +  
  theme_classic()
```

Output:



Scatter plot showing how stress changes with time away from social media.

Interpretation:

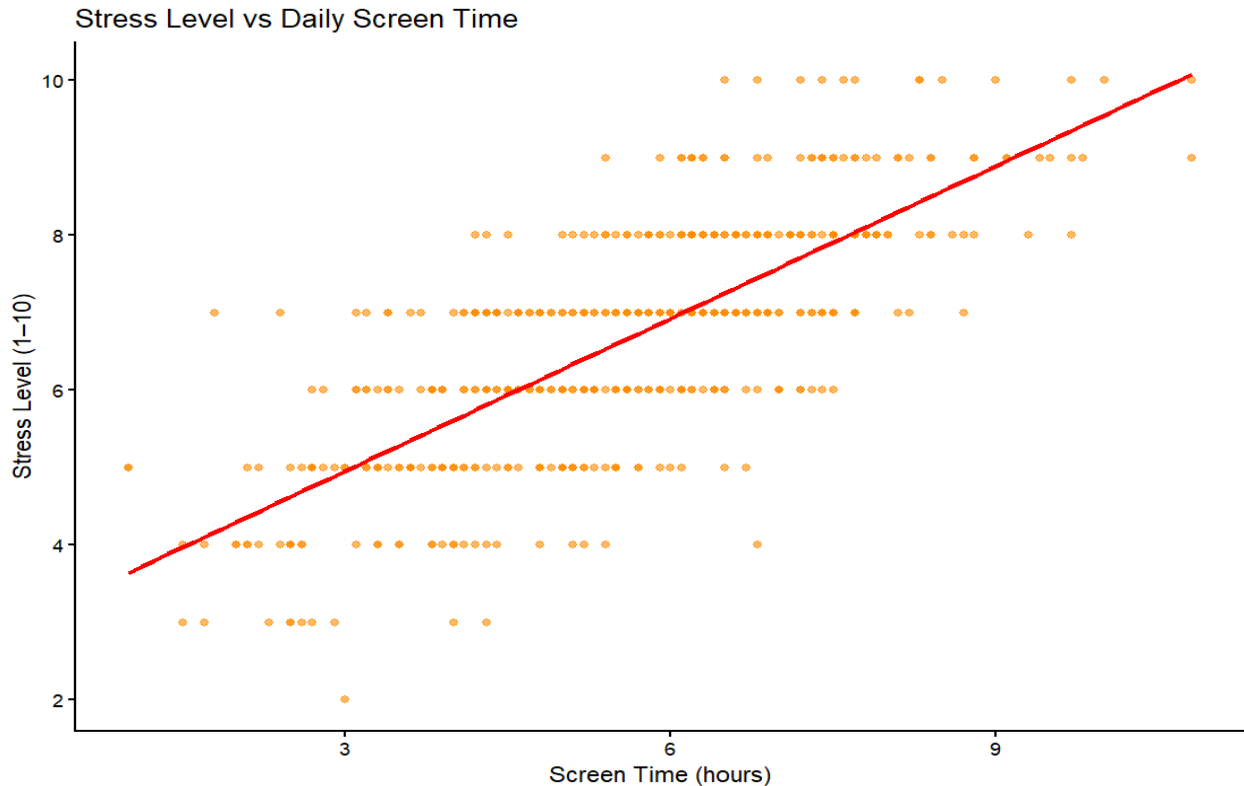
If the trend line slopes downward, it suggests that taking breaks from social media may reduce stress — a valuable insight for mental health recommendations.

Days Without Social Media vs Stress Level:

Code:

```
ggplot(MentalHealth, aes(x = Daily_Screen_Time, y = Stress_Level)) +  
  geom_point(color = "darkorange", alpha = 0.6) +  
  geom_smooth(method = "lm", se = FALSE, color = "red") +  
  labs(title = "Stress Level vs Daily Screen Time",  
        x = "Screen Time (hours)",  
        y = "Stress Level (1–10)" +  
  theme_classic()
```

Output:



Scatter plot showing how stress changes with time away from social media.

Interpretation:

If the trend line slopes downward, it suggests that taking breaks from social media may reduce stress — a valuable insight for mental health recommendations.

Happiness Index by Gender:

Code:

```
ggplot(MentalHealth, aes(x = Gender, y = Happiness_Index, fill = Gender)) +  
  geom_boxplot() +  
  labs(title = "Happiness Index by Gender",  
        x = "Gender",  
        y = "Happiness Index (1-10)") +  
  theme_minimal()
```

Output:**Interpretation:**

- If one gender has a higher median line, it suggests that group reports greater happiness on average.
- If the boxes are taller, that group has more variability in happiness.
- For example, if females show a higher median and tighter box, it suggests more consistent and higher happiness compared to males or others.

